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Participatory action planning as transductive reasoning: towards the right to the city in Los Arenales, Antofagasta, Chile

Francisco Vergara-Perucich *

Abstract This article discusses the experience of creating an overall plan for developing an informal settlement in the Chilean city of Antofagasta. This was accomplished by applying transductive reasoning to a community urban planning practice that aims to advance the community towards achieving the right to the city. By doing so, the article elaborates on insights gathered from the application of transductive reasoning as a method developed by Henri Lefebvre to reflect on the possible futures of cities using a bottom-up approach, thus utilizing the right to the city as a political agenda. Thus, this intervention provides an empirical reflection on the scope of action research in which the ideas of Henri Lefebvre are suitable for transforming utopian thinking into a political agenda for social transformation by a grassroots organization. The conclusions express the critiques of other authors whose approaches to Henri Lefebvre's methods are substantially theoretical, thus wasting its empirical potential and capacity to engage communities.

Introduction

Los Arenales is an informal settlement that is composed mostly of Latin-American migrants who arrived in the city of Antofagasta looking to make a living. The Antofagasta region is known worldwide for its mining industry and high salaries. However, living costs are also high. Since the copper price

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boom ended in 2014, the number of people living in slums has increased by four times according to official surveys. Los Arenales is the biggest slum in the city, is composed of ten campamentos (slum organizations), and contains nearly 1000 households on the northern side of the city, at the foothill. Its people are organized and committed to their struggle for the right to the city. The right to the city is a political agenda that they created after a series of workshops in which the researcher participated. They asked two things of me: first, to help them to better understand what the right to the city entails, and, second, to assist them through the process of collective thinking to devise an idea of how they want to live together.

The initial stage of this process concluded in 2017. In 2018, the community applied for funds to finance modules and activities that would allow them to enhance their education about how to become a self-sufficient community. This paper was written upon the request of community leaders, who encourage lecturers to share their experiences with the world so that their struggle can be visualized and so that the virtues and failings of community engagement using the right to the city as a political framework can be shared.

In practice, to accomplish the community's goals, the process was framed as a participatory action research work through transductive reasoning related to the process of learning urban design based on the right to the city. Transductive reasoning is a research approach that combines inductive and deductive methods to develop a theoretical objective in the form of a hypothesis about a future reality, a utopia (or dystopia) of a possible society. In this case, developing the concept of a concrete utopia was part of the collective production of a plan to transform an informal settlement into an urban space. Specifically, the initiative presented here was part of a formative module funded by the Latin American Know Your City (KYC) programme of Slum Dwellers International (SDI). As the people of Los Arenales were pursuing a way to self-manage their space, the scheme offered by the SDI was suitable. Therefore, a group of community leaders asked various scholars to assist in the formative process by designing a series of modules that would increase the community members' capabilities to integrate slum upgrading projects and prepare for the struggle for their right to the city. The modules focused on cooperativism, human rights, power, leadership, public policy, and the right to the city. The module about the right to the city was inspired by Mark Purcell's approach to the right to the city:

'The right to the city involves two principal rights for urban inhabitants: the right to participation, and the right to appropriation. The right to participation maintains that citizens are inhabitants. The right to participation maintains that citizens should play a central role in any decision that contributes to the production of urban space... The decision could

fall under the auspices of the state (such as a policy decision), of capital (an investment/disinvestment decision), a multilateral institution (a WTO trade ruling), or any other entity that affects the production of space in a particular city. Moreover, the same should be true regarding the tuition (a WTO trade ruling), or any other entity that affects the production of space in a particular city. Moreover, the decision could be made on a range of scales and could involve any level of the state (national, provincial, local), or corporations that operate at any scale (global, national, local)'.

(Purcell, 2002, p. 102).

Purcell's definition is relevant to this research because Los Arenales' *citadins* played a central role in making decisions, having the auspice of an international organization as SDI. This paper presents some reflections from the module on the right to the city, which was based on the transductive approach previously mentioned and which aimed to validate the knowledge of the people of Los Arenales, with emphasis given to their aspirations and wishes regarding how they want to live together. The module consisted of an outline of desired urban planning outcomes (master plan). This master plan was a result of a deliberative contestation on the slum dwellers' conditions in relation to the right to the city principles, which were situated on the map of Los Arenales and defined functions and spaces that helped the people to imagine how their settlement could become an exemplary neighbourhood. Therefore, this paper is a form of participatory action research and contributes to the literature by providing an empirical-based reflection on the scope of transductive reasoning of action. In this way, this study demonstrates the suitability of the ideas of Henri Lefebvre for encouraging a grassroots organization to form a political agenda for social transformation through utopian thinking. As such, this work aims to spark discussions on how transductive reasoning may become a central strategy for bottom-up urban design processes that can be used in any community, including informal communities. Also, the paper contributes to a long tradition of studies on informal settlements developed in Chile.

The study of slum dwellers' organizations in Chile that are looking to attain autonomous urban processes started with a case in La Victoria. In this area, the people claimed a plot of land for building a gigantic neighbourhood that started as slums and later became formalized thanks to community empowerment (Cortés, 2014). Various research works have outlined how people living in *campamentos* formulate strategies to support their livelihood based on self-management and their recognition of the limitations of the political arena in which they work (Valenzuela, Suarez, and Downey, 2010). Under a strict neoliberal regime, the politicization of slum dwellers is vital for them to gain public influence in relation to alleviating their struggles

for the right to housing (Pérez, 2017). Also, they have demonstrated that they are fully capable of managing and determining their own spatial organization (Castillo-Couve, 2014). Los Arenales is composed mostly of Latin-American immigrants living in the extremely impoverished conditions of the Atacama Desert and against the resistance of the authorities and the other citizens of Antofagasta.

This paper presents some reflections on the module that promoted the community's right to the city, which was based on the transductive approach as described previously. The module consisted of a description of an urban planning outcome (i.e. a master plan) as a result of discussions among community members regarding their living conditions in relation to the principles of the right to the city. A map of Los Arenales was drawn up which indicated the functions and spaces that helped the people to imagine how their proposed settlement may become an exemplary neighbourhood. This community process provided the information for the analysis here presented. The data were collected as records and registers during the workshop while building the master plan and discussing decisions. Then, the audio recording was reviewed for the process of coding the original ideas discussed and identifying which information was the most relevant. Therefore, this paper is a form of participatory action research and contributes to the literature by providing an empirical-based reflection on the scope of transductive reasoning of action. In this way, this study presents the suitability of the ideas of Henri Lefebvre for encouraging a grassroots organization to form a political agenda for social transformation through utopian thinking. As such, this work aims to spark discussions on how transductive reasoning could become a central strategy for bottom-up urban design processes that can be used in any community, including informal communities.

The conclusions presented here express a critique of other authors whose approaches to Henri Lefebvre's methods are substantially theoretical, thus wasting its empirical potential and capacity to engage communities through his ideas. In this way, the present article opens a discussion about the role of academia in the production of social alleviation. In this intervention, I call for academia to become more involved in bottom-up processes related to social practices. From my disciplinary field of urban design, I put forward the notion that practitioners should work directly with people as partners rather than considering them to be beneficiaries. Hence, this paper stresses the value of the method employed rather than the outcomes. I made this choice to share this experience with other people working with communities to upgrade slums.

Thus, this article examines the positive outcomes that could emerge from producing collective reflections that ignite the development of new

approaches to their scarcity in a socially embedded utopian way of thinking. As a key contribution to the debate, the involvement of academia as *arrière-garde* thinkers, as opposed to *avant-garde* thinkers, helps to empower grassroots ideas and allows academia to rethink the future based on possibilities rather than on urgent necessities. This way of thinking fosters the development of a long-term common agenda for transforming the current reality. In my view, Henri Lefebvre's concept of transductive reasoning may help to bridge the theory embedded in academic environments with the practice of social transformations.

Methodology: participatory action research as transductive reasoning

The research was designed under the methodological framework of participatory action research. This approach emphasizes the collaborative process of knowledge production through participating in specific activities that trigger profound reflections about certain topics. This approach allows the researcher to investigate sustainable liberation processes for people based on their environment and their engagement in politics through popular education. This approach also promotes outcomes that are derived from the collective process of art production and the reflective practice of performing the actions described above as a community with common goals and aspirations (Barndt, 2011). This methodological approach supports collaborative research in horizontal frameworks in which communities and researchers learn and practice specific activities together that lead towards social change (Kindon, Pain, and Kesby, 2007). Participatory action research allows researchers to address questions and issues that are relevant to decisions that promote social transformation (Reason and Bradbury, 2008). These outcomes are especially important for people living in informal settlements, as is the case for the people of Los Arenales. The goal in the present work was to create an urban image, an idea of the future of Los Arenales as part of a city that currently excludes the community. We employed transductive reasoning to realize a spatial outcome as a response to relevant questions.

In 1970, Henri Lefebvre published *La Revolution Urbaine* (*The Urban Revolution*), a book that integrated most of his reflections on urban questions, ranging from the critique of everyday life to the right to the city. Haunted by the way capitalism was affecting social relations in urban areas, Lefebvre wrote *The Urban Revolution* as a diagnosis and a proposal for radical change in which the praxis of urbanisms occupied a vital role in the difficult task of unleashing urban society (our times) from the profit-oriented rule of capitalism. Provocatively, *The Urban Revolution* presents diverse proposals that Lefebvre organized for unraveling the contradictions of urban

discipline in a society controlled by a capitalist hegemony that may lead towards the emancipation of urban design, architecture, urban planning, geography, and sociology. Thus, the work discusses the development of new theories of urban society. In a way, *The Urban Revolution* constitutes a set of methodological strategies to overthrow capitalism from urban disciplines.

Diverse methodological strategies make *The Urban Revolution* a rich source of urban critiques. To illustrate some contributions, Lefebvre develops the idea of blind fields as a set of social constructs, as spectral walls of division between what is allowed by the hegemonic class and the possibility for one to actually do what is permitted. This division hinders the people's ability to think beyond the limits of the capitalist reality. Similarly, Lefebvre elaborates the idea of urban illusions, which are urban life discourses manufactured from an ideological understanding of everyday life and directed towards making it more efficient for capitalist objectives (control and profit). In relation to the praxis of urban disciplines, Lefebvre also organized the whole book as a comprehensive critique of urban practices. The innovative way that Lefebvre analysed urban practices was theorized by Mark Purcell in relation to democratizing cities (Purcell, 2013), Peter Marcuse regarding social housing as an engine for struggle (Maden and Marcuse, 2008; Marcuse, 2010), Lukasz Stanek in terms of researching architecture as the work of Stanek (2011), to mention just a few scholars who echoed Lefebvre's call to rethink capitalistic urban practices. Indeed, the list of 'Lefebvrians' is long in this regard (Allen and Pryke, 1994; Brenner, 2014; Coleman, 2015; Elden, 2004; Goonewardena *et al.*, 2008; Harvey, 2014; Shields, 2005; Soja, 1980). In this intervention, I would like to present my interpretations and reflections on one particular contribution of Henri Lefebvre's framework: the transductive reasoning.

Henri Lefebvre observed that spatial disciplines recurrently use imagination to design a possible future. He named these intellectual exercises 'concrete utopias' and defined these as hypotheses about urban phenomena that start with the recognition of the factual possibilities for change by embracing the technical, scientific, and social advancements that can create a future social space that is different from the current one. These were considered utopias because they were ideas of a much better future and were labelled 'concrete' because they were grounded in reality. In Lefebvre's interpretation, urban disciplines simultaneously analysed the nature of cities and found ways to make them better. Therefore, Lefebvre presents transduction as a type of analysis that produces a diagnosis with transformative outcomes from research but in which the outcome is already present from the beginning of the formulation of the research in the form of a hypothesis, an imagined future or a virtual object of the city. By

elaborating transductive reasoning, Lefebvre contributed to urban studies by implementing a scientific approach to utopian thinking. Mark Purcell explains that transductive reasoning is 'a way to cut a path that leads beyond the actual world already realized and toward a possible world yet to come' (Purcell, 2013, p. 21).

In *The Urban Revolution*, Lefebvre introduces transductive reasoning as the leading criterion for the organization of his book. For instance, the idea of an urban society as a virtual object emerged from the transductive reasoning that sustains the ongoing process of the replacement of an industrialized capitalism by an urbanized capitalism. For Lefebvre, such processes were also possibilities to revolutionize everyday life and separate capitalism from urban practices. The urban society was indeed a possible future, though not yet realized, such that diverse changes could redirect its aims from capitalism to a humanized urban life by using tools such as the right to the city as a map of a pursued future. Thus, the urban society is also a political instrument of research that fosters the development of purposeful conclusions. Transductive reasoning forces researchers to abandon a neutral political stance and makes them choose a side.

Furthermore, transduction may serve to research and transform social reality through articulating theory and practice, ideas with facts and interpretations with actions. Transduction employs a dialectical strategy to confront different approaches to interpreting reality. This strategy commits to the development of revolutionary reflections because it aims to promote changes that go beyond the limits imposed by the blind fields of capitalism. As put forth in most of the work of Henri Lefebvre, transductive reasoning is not prescriptive. Instead, Lefebvre opens a methodological discussion on how to conduct urban research by illuminating contradictions between theory and practice, method and methodology. In his own words, transduction 'is an intellectual operation which can be methodically carried out and which differs from classical induction, deduction, the construction of "models" simulation as well as the simple statement of the hypothesis' (Lefebvre, 1996, p. 151). Although it is close to being a speculative exercise, transduction uses information to elaborate its initial theoretical reflections that then need to be tested in a continuous cycle of feedback between theoretical constructs and empirical observations.

In the approach developed with the dwellers of Los Arenales, the use of transductive reasoning was tested by creating a clear notion of a concrete utopia using the methodology of participatory action research as a framework. The people of Los Arenales wanted to upgrade their living conditions based on a right to the city agenda, and we collaboratively developed a master plan by applying transductive reasoning to advance their political

agenda. Thus, the theory of the right to the city exposes the contradictions of empirical reality. In doing so, this case offered a unique opportunity to test the suitability of transductive reasoning in practice to implement urgent changes in a community that was living in scarcity. As Marx stated in the *Thesis on Feuerbach*, 'The philosophers have only interpreted the world, in various ways; the point is to change it' (Marx, 1976, p. 65). In contributing to this challenge, the article valorizes the role of urban design practitioners in advancing critical thinking among communities and helping them to understand the importance of spatial practice in their everyday struggles (Figure 1).

The case study: Macrocampamento Los Arenales, Antofagasta, Chile

In 2010, Chile was invited to become an OECD member. Since then, the nation has contributed as one of the leading economies in Latin America to this select group. However, macroeconomics does not necessarily fit with local realities. Indeed, the Chilean dependence on minerals and raw materials has characterized its history of inequality (Salazar, 2003). As a country that is highly dependent on mining activities, its principal extractive is in the northern region of Antofagasta, which contains 13 percent of the country's mining activities and which has the largest amount of copper in the world (Arias *et al.*, 2014). This region's capital city is also named Antofagasta, and it is the most populous metropolis in northern Chile – for check the location see Figure 2. Its mineral dependency has produced more than a few economic bonanzas and consequent socio-economic crises because of the fluctuant nature of commodities in financial markets. Nevertheless, the economic growth of this region might not have increased the value of regional industrial processes, and the Chilean economy has employed non-sustainable ways of keeping the mining industries alive (Phelps *et al.*, 2015), which has undermined the stability of the commodity regions, such as Antofagasta.

Still, the Chilean productive fabric and the impressive recent macroeconomic growth figures have been based on extractive industries, especially mining. This mono-dependency was initiated with the neoliberal policies that took place during the dictatorship and has been deepened by every successive democratic government after the return to democracy in 1990. This pattern has produced economic growth for the nation and its extractive regions, but not in a sustainable way. In 2011, Chile doubled its 1988 GDP mostly because of the boom in the international copper price (OECD, 2013), which led to high salaries throughout the boom period (i.e. until 2013).

This increase in salaries was reflected in cities such as Antofagasta, where the cost of life went up as the nation's population migrated to Antofagasta

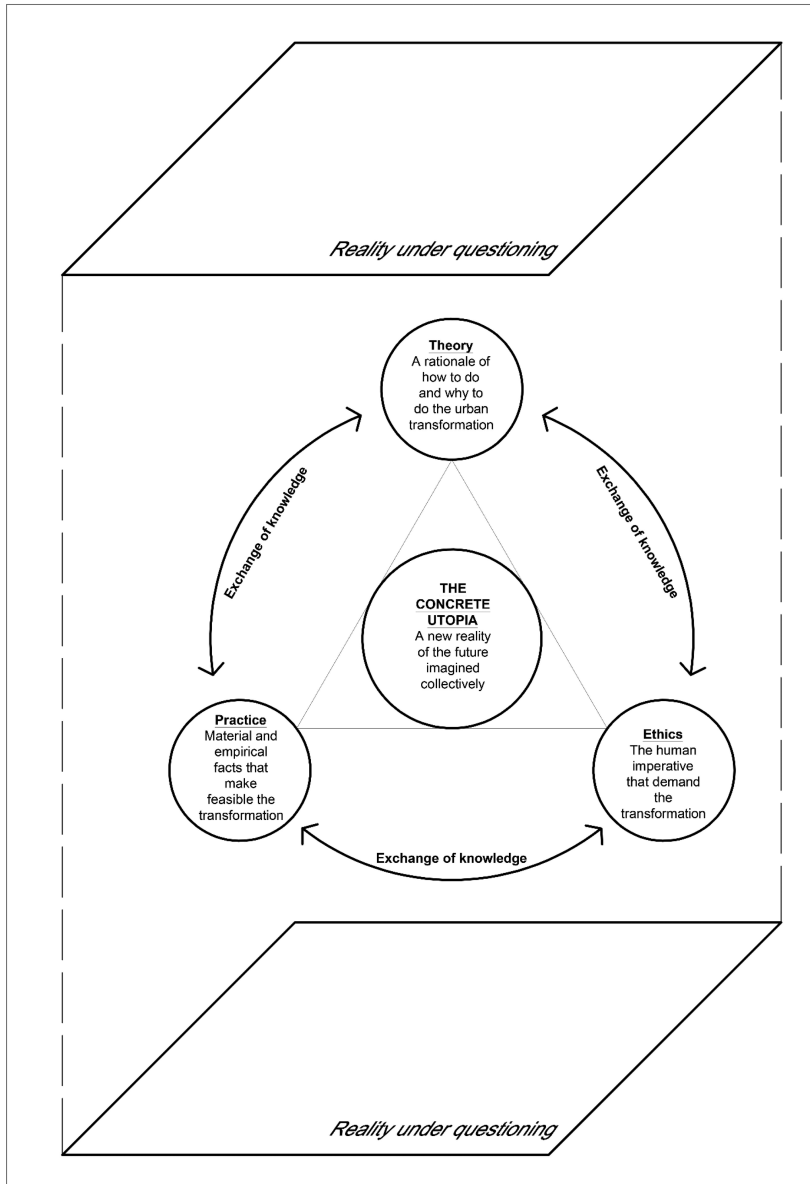


Figure 1 A synthesis of transductive reasoning for the case of this research. Source: own elaboration. This figure illustrates the process of transductive reasoning. At the center of the figure is the concrete utopia (the virtual object, the possible future of a reality that may be explored by an empirical-theoretical approach towards changing certain ethos or praxis in reality). By questioning this object ethically, practically, and theoretically, a series of contradictions is revealed regarding the relationship between the parts that provide a critical understanding of the possibilities, obstacles, conflicts, and virtues that may come with the transformation of space



Figure 2 Geographical location of Los Arenales. Source: author based on [Instituto Nacional de Estadísticas \(2018\)](#)



Figure 3 Picture of Antofagasta's foothills with slums built along them. Source: author

to find jobs in the global capital of copper mining. Sadly, the boom ended around 2013, which led to a severe crisis in the urban life of Antofagasta. After Santiago, Antofagasta is the most expensive Chilean city to live in (Paredes *et al.*, 2016), and the city's unsustainable economic model makes it highly vulnerable to mining cycles and international commodity pricing (Phelps *et al.*, 2018). Since the bonanza years, the city has faced two striking problems: the high international migration of people from other Latin American countries who are looking for the job opportunities fueled by mining industries (whose bonanza is not booming anymore) and the drastic increase in the number of informal settlements (*campamentos*) that are reshaping the area in the foothills of the city – as in the hills showed in Figure 3. The last census of 2017 indicated that 13 percent of people living the city are foreign immigrants (Instituto Nacional de Estadísticas, 2018). Between 2005 and 2014, immigration grew by 327 percent (Departamento de Extranjería y Migración, 2015). Furthermore, the CENSUS 2017 indicated that most of the foreign population in Antofagasta was comprised of Colombians, Bolivians, and Peruvians and that 12 percent of these foreigners are living in poverty. These data may explain the 971 percent increase in the number of families living in informal settlements in Antofagasta from 2007 to 2017 (TECHO, 2018). This is an extremely high rate for a city in a country that is supposedly one of the 37 most advanced economies in the world.

Nationally, the number of families living in *campamentos* increased by 42 percent between 2011 and 2016. This situation may be connected with the national increase in housing prices, which reached a peak of 57 percent from during these years,¹ while the consumer price index rose by 33.8 percent during the same period.² This variation is especially visible in the north of Chile, where housing prices increased by 82 percent between 2011 and 2016.³ According to a Paulo Flores, the number of families living in *campamentos* increased by 40,541, possibly as a consequence of the low affordability of living in the rental sector of the housing market (Flores, 2016). For instance, among all OECD country members, household expenditures make up about 21 percent of the average families living costs, while families in Chile allocate 34.5 percent of their net incomes to housing.

Indeed, life is particularly difficult for families in the more impoverished quintile (20 percent) of the population, who spend 78 percent of their household net income on average on rent (Ministerio de Desarrollo Social, 2015). To add more complexity to the issues of those living in *campamentos* in

1 Source: Banco Central de Chile (Central Bank) statistics about housing prices, 2017.

2 Source: Instituto Nacional de Estadísticas (National Institute of Statistics), 2017.

3 Source: Banco Central de Chile (Central Bank) statistics about housing prices, 2017.

Antofagasta, data from the central government (GORE, 2016) indicate that 60 percent of people living in Antofagasta's *campamentos* are foreigners. Given the high prices of housing, living in *campamentos* has become a survival strategy for those wishing to stay near this urban centrality while they wait for another boom in copper prices.

One of the most complex *campamentos* in the city is Los Arenales. It is a vast territorial unit composed of around 1000 houses that are organized into thirteen different *campamentos*, all unified under a central board of democratically elected representatives. As is usual in Antofagasta, Los Arenales occupies the foothills of a publicly owned plot of land (20 hectares) in the northeast side of the city. The soil is composed mostly of sand ('Arenales' means 'sandbank' in English). The soil is irregular, dry and lacks vegetation. Urbanization is completely absent in the area. Most of the houses are built from sheets of wood and zinc.

The internal geopolitics is no less complex than the environment in which they are placed. Around 80 percent of Los Arenales's dwellers are immigrants (GORE, 2016). United by cultural differences and their alien status, the people of Los Arenales have come to embrace and value their diversity. After a couple of years of collective organization, they managed to create a sense of belonging in the informal neighbourhood that they aimed to transform from shanty towns to places where the otherness and dignity of the poor can be respected. This is their very own concrete utopia which sets an example for other informal settlements in Latin America. The people of Los Arenales have developed political skills to create networks with strategic partners such as NGOs, local authorities, private companies and academia. Also, they have fostered an informal but intense academic formation for building the capacities to theorize about and analyse their problems. All these efforts are organized under a political agenda of the right to the city.

In April 2018, Los Arenales received funding from SDI through their programme, KYC Latin America. Parts of their project included a participatory mapping process, a census, and an educational course for building capabilities on certain topics, such as human rights, inequality, power, history, urban policy, migration, solidary economics, gender, and the right to the city. The following part of this paper presents the results of the module of the right to the city based on transductive reasoning and the collective production of a concrete utopia for Los Arenales which empowers the dwellers by urban design.

Applying transductive reasoning towards the right to the city in Los Arenales

In July 2018, architect Julio Santander – who is also the regional secretary for the Housing and Urbanism Ministry – said to the dwellers of Los Arenales

that the *campamentos* were suitable for settling in the same place. His words were important to the dwellers, given that, so far, they had constantly been worrying that the police would come to evict them because their occupation of the area was illegal. The words of Santander calmed the people and fed their hope of finding a solution to their problem and fostering conviction in their struggle for the right to the city (though this struggle had started some time before this).

In April 2017, Leilani Farha, the special rapporteur on adequate housing of the United Nations, visited Los Arenales to investigate settlements in Chile. Farha's conclusions were categorical: 'Inadequate urban planning and zoning have given more weight to land regarding profit, rather than regarding its social functions or its centrality for residential housing' (Farha, 2018, p. 9). Since then, the Observatorio Regional de Desarrollo Humano (ORDHUM) of Universidad Catolica del Norte was invited to give lectures to teach the dwellers of Los Arenales about the right to the city. These sessions covered both the theory and practice of self-sufficient urban development in a political agenda for transforming the city based on the dwellers' desires and aspirations (Harvey, 2008; Lefebvre, 1969), aiming to produce a good city from the grassroots (Friedmann, 2000). The primary goal of this series of lectures was not only to present the theory of the right to the city but also make the dwellers believe that they can be engines of social change for good.

Four sessions were held: (i) the first was based on the concerns and reasons that make this group of people live in an informal settlement; (ii) the second session for presenting the theory and history of the right to the city from the Paris Commune to the May 1968 in Paris, with Lefebvre as the central figure of the session; and (iii) the third session aimed to situate Los Arenales in the broader context of struggles around the world for the right to the city. These three sessions served to build a set of knowledge and ideas that aimed to open the minds of dwellers to new possibilities for the future based on the realization that similar struggles have produced significant changes in another part of the world. Using a transductive rationale, these three sessions constructed a theory of why it was important for the people of Los Arenales to transform their urban space and that it is an ethical matter. The dwellers were made aware of why they were pushed outside the city to live in informality: 'I did not have enough money for paying the rent; it was too much' (Woman, 50–60 years old).

Certainly, between 2000 and 2017, the average price of renting housing in Antofagasta increased by 538 percent, while incomes increased by only 129 percent (MIDESO, 2018a). This was a material dimension of the problem, see Table 1, in which having access to urban life becomes unaffordable because of a series of neoliberal policies that deregulate the

Table 1 Variation of household income, renting prices, and ratio measuring the weight of renting prices in household income in the city of Antofagasta

Year	Average household income (UF)	Average renting prices (UF)	Renting/income
2000	17.57	1.55	9%
2003	36.74	2.36	6%
2006	46.00	2.37	5%
2009	41.99	2.41	6%
2011	53.98	9.91	18%
2013	50.25	10.57	21%
2015	44.61	9.41	21%
2017	40.23	9.89	25%

Source: author based on [MIDESO, 2018a, b](#)

housing market and rely too much on free-market economics ([Boano and Vergara-Perucich, 2017](#)). Furthermore, Antofagasta's quantitative housing deficit is 22,260 units ([MIDESO, 2018b](#)), which makes people living in scarcity more exposed to the dark side of this deficit being expelled from formality to live in the open field of one of the driest deserts in the world.

'I'd rather live here, in the dust and under the heavy sun, than in the city where someone is taking advantages of my condition as migrant and poverty, at least here I choose how to live' (Man, 30–40 years old).

These ethical questions have emerged from the people of Los Arenales who served to then build a political agenda of the right to the city. This agenda was against the profit-oriented logic of neoliberal urban development ([Arboleda, 2016](#)) and fostered a more humane way of developing the space ([Harvey, 2014](#)). The fourth session, the one that is reported here, focused on the practice and the material reflections about the future of Los Arenales by developing a concrete utopia, a new reality for the *campamento* imagined collectively and aimed at materializing the theoretical and ethical reflections in a visualized possibility of the future.

As discussed previously, the exchange between theory, practice, and emotions is vital for creating a concrete utopia. To accomplish this took over 16 months of working with the people of Los Arenales about the right to the city. This process of building an understanding of the right to the city from the particular context of an informal settlement placed in a northern city in Chile is quite complex. Yet it is crucial for realizing how ideas that originated far away (in Paris in this case) can be adopted by people who need these ideas, too in order to create a grounded version based on cultural values and specific needs. In this case, the multicultural composition of Los Arenales and its population's need for housing were the

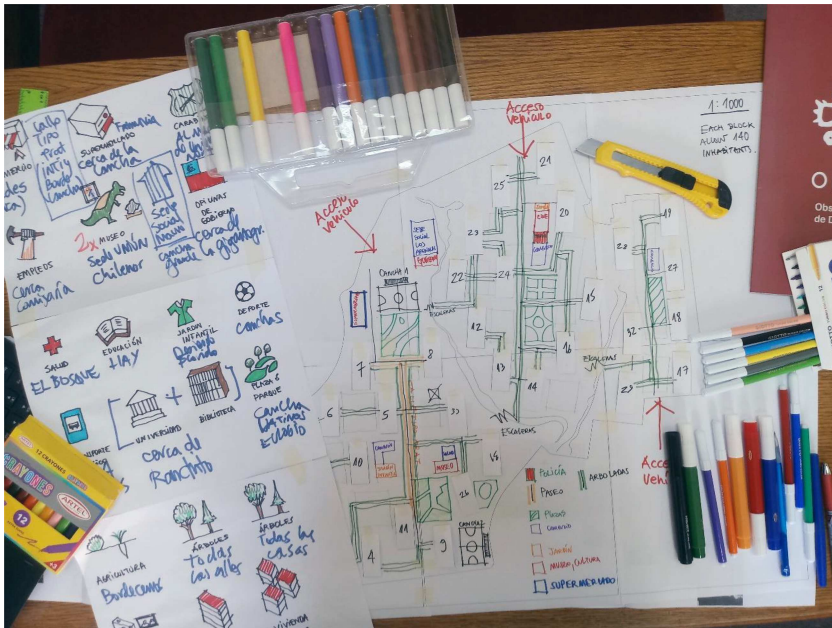


Figure 4 Picture of the raw materials employed in the participatory process of elaborating the master plan for the concrete utopia of Los Arenales. Source: author in collaboration with people from Los Arenales

main engines of creating a theory for the right to the city rooted in their own interpretation.

‘This city, Los Arenales, has been built with our bare hands, something impossible in the regular city. Ok, this is not the most beautiful area of the city but at least is ours’ (Woman, 40–50 years).

Following the transductive method, the dwellers came to believe in their dream of a better place to live based on their interests. Thus, under the scheme of the right to the city (and being that this was the final session), the dwellers called for imagining the concrete utopia. They wanted to see what it would look like if all these ideas and theories become true. As part of a collective spatial reflection, we produced a series of demands for this new space, indicating which facilities should be located in which areas of Los Arenales.

From a scratched first version resultant from the participatory design, as shown in Figure 4, we moved to a more formally designed plan for Los Arenales, see Figure 5. This ‘master plan’ emerged from a series of discussions and disputes about the definition of what a concrete utopia was for these people. As such, the result is also a representation of these social relations (Lefebvre, 1991). The result is not too different from any other kind of master plan; the difference is the process. People designed this

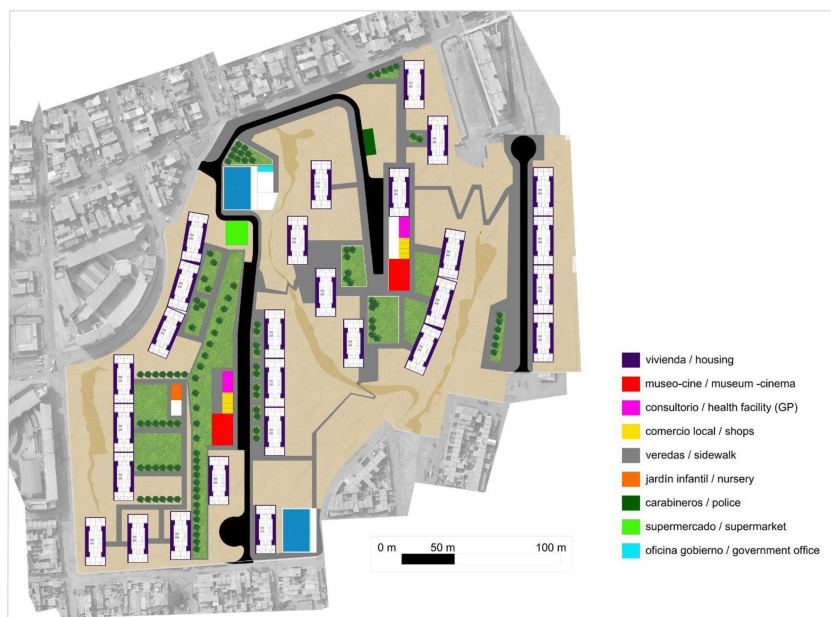


Figure 5 Master Plan of the Concrete Utopia for Los Arenales. Source: author in collaboration with people from Los Arenales

master plan, decided where to put each service and facility, and prioritized spaces based on their own perspectives and desires. Most of those who participated in this activity wanted to build four-story collective housing buildings, each with a capacity of 150 people. They demanded to have local shops, a small supermarket, cultural buildings, a police station, governmental services, sports and recreation facilities, bus stops, a nursery, a higher education branch, trees, squares, and a central headquarters for community meetings (a sort of Parliament of Los Arenales) for managing internal governance. For evaluating the feasibility of this concrete utopia, it was required to quantify the budget required to its implementation (Table 2).

The master plan has a capacity of around 4500 inhabitants. In order to quantify the costs of this concrete utopia, its construction would imply an investment of USD 12,192,282 per year over 5 years. This amount of money represents 5.09 percent of the annual public funds available in the region for urban development, see Table 2. The benefit of this project is that it assists with the housing deficit in 8.2 percent of the whole demand, and reducing a 49.27 percent of the household living in overcrowded conditions. Also, the project increases the presence of services in an isolated area of the city. For the dwellers who participated in this transductive initiative the utopian reflections and spatial proposals seems distant and yet urgent.

Table 2 Resources available per public service entitled to invest in urban development for low-income communities, project items, and its costs and the quantification of the yearly cost of this project in a 5-year basis of implementation

Resources	USD
Municipal Budget for Public Works in Antofagasta	6,615,384
National Solidary Funds Subsidies Available for Antofagasta per Year	110,334,000
National Fund for Regional Development (FNDR)	122,548,603
Annual Regional Budget for Public Works	239,497,987
Project items per investment	
Urbanizing the plot of Land	1,076,923
Police Station	452,489
Museum-Cinema	615,385
Health Facility	2,307,692
Nursery	1,153,846
Government office	307,692
Parliament of Los Arenales	615,385
Housing Construction	54,432,000
Total Budget Required for Concrete Utopia in Los Arenales	60,961,412
Yearly investment	12,192,282
Portion of the Annual Regional Budget for Public Works	5.09%

Sources: author based on SUBDERE (2018, 2017); MINVU (2018); ChileCompra (2018)

Conclusions

Conducting participatory action research based on transductive reasoning allows researchers to explore the spatial products that lead towards the materialization of the ideas related to a community's future. It provides both an image of a future space and a collective deliberative process for making spatial decisions that shape that space. In this case, the master plan represents those decisions and is composed by the community with the assistance of experts. On the one hand, the master plan has technical value for discussing with the authorities how the people of Los Arenales want to live in the future. On the other hand, it is a record that explains how the space can be organized collaboratively. The use of transductive reasoning in participatory action research could guide urban planning processes within a grassroots society, not as a pragmatic approach to housing needs but as an imaginative exploration of what is possible. In Los Arenales, imaginative exploration entails the visualization of a community in which the right to the city has been achieved. By knowing the desired outcome, the path to that outcome is easier to track.

Transductive reasoning is a process of grassroots urban design which has several positive outcomes: (i) it invites dwellers living in a precarious situation to imagine a better future, to embrace the possibility of making it real, and to valorize each step towards the utopian idea of living better (which, in reality, is not so utopian); (ii) as part of the efforts towards that

aim, dwellers receive training to increase their capabilities to understand their context and make them more aware of the complexities that they will face under such specific conditions; and finally (iii), it serves to test how suitable revolutionary theories – such as the right to the city or the concrete utopia – are for studying and performing scholar activism in reality. Many scholar activists prefer a pragmatic approach to social change which deludes people about the possibility of reflecting collectively in less realistic scenarios, thus inviting people to think beyond the borders of neoliberal blind fields. In my experience, this has been an enlightening way to see how people can afford to dream at the same time that they learn how to deal with complexities in reality. The concrete utopia, as an outcome of transductive reasoning, still has further potential that may require further explorations using different approaches in different contexts.

In Chile, the average time a person lives in *campamentos* is around 8 years. This is a long time, often resulting in the person abandoning the possibility of developing further skills as they become motivated only by urgencies and immediate needs. The city is constantly in movement, and so too should the minds of people living in scarcity. In the 1960s, Henri Lefebvre invited a whole generation to dream of a different future, and they did. In a way, they reshaped their futures. Today, under a neoliberal regime, the tools and provocations developed by Lefebvre still are useful for imagining a better urban life. It was refreshing to see that the community leaders connected with the ideas of Lefebvre and for them the transductive reasoning made sense to foster a collective sense of building the space together.

As a final conclusion, I would like to invite to other authors whose approach to Henri Lefebvre's methods is substantially theoretical (as it used to be my own) to challenge communities by using his ideas and to lead the people of these communities to believe that it is possible to hold an open discussion about the role of the concrete utopia in changing reality, especially for the dispossessed and displaced. From my disciplinary field of urban design, I wanted to share this experience to show some positive outcomes that may emerge when producing collective reflections that ignite a collective utopian way of thinking. Maybe this community would not concretize their utopia, but at least now they have more to fight for: a dreamed image of their own city based on their own principles, desires, and aspirations.

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References

- Allen, J. & Pryke, M. (1994). The production of service space. *Environment and Planning D: Society and Space*, **12** (4), 453–475. <https://doi.org/10.1068/d120453>.
- Arboleda, M. (2016) Spaces of extraction, metropolitan explosions: planetary urbanization and the commodity boom in Latin America, *International Journal of Urban and Regional Research*, **40** (1), 96–112.
- Arias, M., Atienza, M., Cademartori, J. (2014) Large mining enterprises and regional development in Chile: between the enclave and cluster, *Journal of Economic Geography*, **14** (1), 73–95.
- Barndt, D. (2011). *!Viva!: Community arts and popular education in the Americas. !Viva!: Community Arts and Popular Education in the Americas*. Albany: State University of New York Press. Retrieved from <http://www.scopus.com/inward/record.url?eid=2-s2.0-84900889711&partnerID=40&md5=a4cb34db731a6e226fa349edbee8511a>.
- Boano, C. and Vergara-Perucich, F. (2017) *Neoliberalism and Urban Development in Latin America*, Routledge, Oxon and New York, NY.
- Brain Valenzuela, I., Prieto Suarez, J. J., Sabatini Downey, F. (2010) Vivir en Campamentos: ¿Camino hacia la vivienda formal o estrategia de localización Para enfrentar la vulnerabilidad? *Eure*, **36** (109), 111–141.
- Brenner, N. (2014). Implosions/Explosions: Towards a Study of Planetary Urbanization: Neil J Brenner: 9783868593174: [Amazon.com](http://www.amazon.com/Implosions-Explosions-Towards-Planetary-Urbanization/dp/3868593179/ref=sr_1_1?ie=UTF8&qid=1434621477&sr=8-1&keywords=implosions+explosions): Books. Retrieved from http://www.amazon.com/Implosions-Explosions-Towards-Planetary-Urbanization/dp/3868593179/ref=sr_1_1?ie=UTF8&qid=1434621477&sr=8-1&keywords=implosions+explosions.
- ChileCompra. 2018. *Mercado Público - La Plataforma de Compras Públicas y Oportunidades de Negocio Del Estado de Chile*, Listado de Compras Publicas. 2018, accessed at: <https://www.mercadopublico.cl/Home> (22 November 2019).
- Coleman, N. (2015). *Lefebvre for Architects*. London - New York: Routledge.
- Cortés, A. (2014) El movimiento de pobladores chilenos y la población La Victoria: ejemplaridad, movimientos sociales y el derecho a la ciudad, *EURE (Santiago)*, **40** (119), 239–260.
- Farha, L. (2018) *Report of the Special Rapporteur on Adequate Housing as a Component of the Right to an Adequatestandard of Living, and on the Right to Non-Discrimination in this Context, on Her Mission to Chile*, Vol. A/HRC/37/5, New York, NY, accessed at: <https://documents-dds-ny.un.org/doc/UNDOC/GEN/G18/009/43/PDF/G1800943.pdf?OpenElement> (15 October 2018).
- Flores, P. (2016) Los Campamentos en el Chile de Hoy, *Revista CIS*, **1** (20), 7–10.
- Friedmann, J. (2000) The good city: in defense of utopian thinking, *International Journal of Urban and Regional Research*, **24** (2), 460–472.
- Goonewardena, K., Kipfer, S., Milgrom, R., & Schmid, C (2008). *Space, difference, everyday life: Reading Henri Lefebvre* (Vol. 65). , London – New York. https://doi.org/10.1111/j.1745-7939.2009.01167_6.x.

- GORE (2016) *Catastro y Encuesta a Familias de Campamentos, Ciudad de Antofagasta, Año 2015*, GORE, Antofagasta.
- Harvey, D. (2008) The right to the City, *New Left Review*, **53** (53), 23–40.
- Harvey, D. (2014) Diecisiete Contradicciones y El Fin Del Capitalismo, *Journal of Chemical Information and Modeling*, **53**. doi: <https://doi.org/10.1017/CBO9781107415324.004>.
- Inmigraciones, departamento de extranjería e. (2015). Estadísticas migratorias del departamento de extranjería y migración 2015. *Departamento de Extranjería y Migración Del Ministerio Del Interior y Seguridad Pública*, **116**. Retrieved from <http://www.extranjeria.gob.cl/media/2017/07/AnuarioEstadisticoNacionalDEM2015.pdf>.
- Instituto Nacional de Estadísticas. (2018). CENSO 2017. Retrieved February 1, 2019, from <https://www.censo2017.cl/>.
- Instituto Nacional de Estadísticas. (2018) CENSO 2017, Censo 2017.
- Kindon, S., Pain, R., & Kesby, M. (2007). *Participatory action research approaches and methods* (Vol. **94**). Routledge, Oxon, New York.
- Lefebvre, H. (1969) *El derecho a la ciudad*. Editorial Península, Barcelona.
- Lefebvre, H. (1991) *The Production of Space*, Blackwell, Oxford and Cambridge.
- Lefebvre, H. (1996) In E. Kofman, E. Lebas eds, *Writing on Cities*, Blackwell, Oxford, MA.
- Maden, D., & Marcuse, P. (2016). *In Defense of Housing*. Verso, London, New York. <https://doi.org/10.1300/J159v03n01>.
- Marx, K. (1976) *Theses on Feuerbach*, Foreign Languages Press, Peking, accessed at: <https://doi.org/10.4324/9780203543375>.
- Marcuse, P. (2010). In defense of theory in practice. *City*, **14** (1), 4–12. <https://doi.org/10.1080/13604810903529126>.
- MIDESO. (2018a) Bases de Datos Históricas Encuesta CASEN, *Encuesta de Caracterización Socioeconómica Nacional*, CASEN, accessed at: <http://observatorio.ministerio.desarrollosocial.gob.cl/casen-multidimensional/casen/basedatos.php> (10 September 2018).
- MIDESO. (2018b) Informe de Desarrollo Social 2018, Santiago, accessed at: http://www.ministeriodesarrollosocial.gob.cl/storage/docs/Informe_de_Desarrollo_Social_2018.pdf (2 December 2018).
- Ministerio de Desarrollo Social. (2015) Encuesta CASEN 2015. Ministerio de Desarrollo Social. Santiago, Chile.
- MINVU. (2018) Estadísticas Habitacionales. Observatorio Urbano, accessed at: <http://www.observatoriourbano.cl/estadisticas-habitacionales/> (2 December 2018).
- OECD. 2013. *OECD Territorial Reviews: Antofagasta, Chile*. OECD, Paris, accessed at: <https://doi.org/10.1787/9789264203914-en>.
- Paredes, D., Iturra, V., Lufin, M. (2016) A spatial decomposition of income inequality in Chile, *Regional Studies*, **50** (5), 771–789.
- Pérez, M. (2017) “A new Poblador is being born”: housing struggles in a gentrified area of Santiago, *Latin American Perspectives*, **44** (3), 28–45.
- Peter REASON, & Hilary BRADBURY. (2008). *SAGE Handbook of Action Research: Participative Inquiry and Practice*. Retrieved from <https://ikhsanaira.files.wordpress.com/2016/09/action-research-participative-inquiry-and-practice-reasonbradburry.pdf>.

- Phelps, N. A., Atienza, M., Arias, M. (2015) Encore for the enclave: the changing nature of the industry enclave with illustrations from the mining industry in Chile, *Economic Geography*, **91** (2), 119–146.
- Phelps, N. A., Atienza, M., Arias, M. (2018) An invitation to the dark side of economic geography, *Environment and Planning A*, **50** (1), 236–244.
- Purcell, M. (2013) *The Down-Deep Delight of Democracy*, John Wiley & Sons, Oxford, MA and West Sussex.
- Salazar, G. (2003) *Historia de La Acumulación Capitalista en Chile (Apuntes de Clases)*, LOM Ediciones, Santiago.
- Shields, R. (2005). *Lefebvre: Love and Struggle. Spatial Dialectics* (Vol. **53**). Routledge, London, New York. <https://doi.org/10.1017/CBO9781107415324.004>.
- Soja, E. W. (1980). The Socio-Spatial Dialectic. *Annals of the Association of American Geographers*. <https://doi.org/10.1111/j.1467-8306.1980.tb01308.x>.
- Stanek, Ł. (2011). *Henri Lefebvre on Space* (Vol. **1**). University of Minnesota Press, Minneapolis, London. <https://doi.org/10.1017/CBO9781107415324.004>.
- SUBDERE (2017) FNDR: Ejecución Presupuestaria Diciembre 2017 | SUBDERE, *Fondo Nacional de Desarrollo Regional*, 2017, accessed at: <http://www.subdere.gov.cl/documentacion/fndr-ejecución-presupuestaria-diciembre-2017>.
- SUBDERE (2018) Sistema Nacional de Información municipal, *Ficha Comunal Antofagasta*, accessed at: http://datos.sinim.gov.cl/ficha_comunal.php (2 December 2018).
- TECHO. (2018) Monitor de Campamentos, Monitor de Campamentos En Chile, accessed at: <http://chile.techo.org/cis/monitor/monitor.php#> (10 September 2018).